

NDA Mathematics — Vectors, Probability & Binomial Distribution

1. The magnitude of the vector $\vec{a} = 3\hat{i} - 2\hat{j} + 6\hat{k}$ is:

- (a) $\sqrt{41}$
- (b) $\sqrt{11}$
- (c) 7
- (d) 11

- (a) $\frac{5}{2}$
- (b) 5
- (c) $\frac{2}{5}$
- (d) $-\frac{5}{2}$

2. A unit vector in the direction of $\vec{a} = 4\hat{i} - 3\hat{j}$ is:

- (a) $\frac{1}{25}(4\hat{i} - 3\hat{j})$
- (b) $\frac{1}{5}(4\hat{i} - 3\hat{j})$
- (c) $\frac{1}{7}(4\hat{i} - 3\hat{j})$
- (d) $4\hat{i} - 3\hat{j}$

10. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and the angle between them is 60° , then $|\vec{a} + \vec{b}|$ equals:

- (a) $\sqrt{13}$
- (b) $\sqrt{37}$
- (c) $\sqrt{25}$
- (d) 7

3. If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, then a vector of magnitude 9 in the direction of $\vec{a}$ is:

- (a) $9\hat{i} + 9\hat{j} + 9\hat{k}$
- (b) $6\hat{i} + 3\hat{j} + 6\hat{k}$
- (c) $2\hat{i} + \hat{j} + 2\hat{k}$
- (d) $3\hat{i} + 3\hat{j} + 3\hat{k}$

11. If $\vec{a}$ and $\vec{b}$ are such that $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 3$, then the angle between them is:

- (a) 60°
- (b) 30°
- (c) 45°
- (d) 90°

4. If $|\vec{a}| = 5$, $|\vec{b}| = 13$ and $|\vec{a} \times \vec{b}| = 25$, then $\vec{a} \cdot \vec{b}$ is equal to (taking the positive value):

- (a) 25
- (b) 65
- (c) 60
- (d) $\sqrt{60}$

12. If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ with $|\vec{a}| = 3$, $|\vec{b}| = 5$ and $|\vec{c}| = 7$, then the angle between $\vec{a}$ and $\vec{b}$ is:

- (a) 120°
- (b) 90°
- (c) 60°
- (d) 45°

5. If $\vec{a}$ and $\vec{b}$ are unit vectors with $|\vec{a} + \vec{b}| = \sqrt{3}$, then $|\vec{a} - \vec{b}|$ equals:

- (a) $\sqrt{3}$
- (b) 2
- (c) 1
- (d) $\sqrt{2}$

13. If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is the angle between them, then $\frac{1}{2}|\vec{a} - \vec{b}|$ is equal to:

- (a) $\tan \frac{\theta}{2}$
- (b) $\sin \frac{\theta}{2}$
- (c) $2 \sin \frac{\theta}{2}$
- (d) $\cos \frac{\theta}{2}$

6. If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$, then $\vec{a} \cdot \vec{b}$ is equal to:

- (a) 3
- (b) 7
- (c) -3
- (d) -1

14. The scalar projection of $\vec{a} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$ is:

- (a) $\frac{20}{3}$
- (b) $\frac{20}{7}$
- (c) 20
- (d) $\frac{10}{3}$

7. The angle between the vectors $\hat{i} + \hat{j}$ and $\hat{j} + \hat{k}$ is:

- (a) 60°
- (b) 45°
- (c) 30°
- (d) 90°

15. The vector projection of $\vec{a} = 3\hat{i} + \hat{j}$ on $\vec{b} = \hat{i} + \hat{j}$ is:

- (a) $\hat{i} + \hat{j}$
- (b) $2\hat{i} + 2\hat{j}$
- (c) $4\hat{i} + 4\hat{j}$
- (d) $2\hat{i} + \hat{j}$

8. If $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} + 4\hat{k}$, then the cosine of the angle between $\vec{a}$ and $\vec{b}$ is:

- (a) $\frac{3}{5}$
- (b) $\frac{7}{15}$
- (c) $\frac{11}{25}$
- (d) $\frac{11}{15}$

16. If the projection of $\vec{a} = \hat{i} + \lambda\hat{j} + \hat{k}$ on $\vec{b} = 2\hat{i} + \hat{j} + 2\hat{k}$ is 2, then λ equals:

- (a) 3
- (b) 2
- (c) -2
- (d) 1

9. If the vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular, then λ equals:

NDA Mathematics — Vectors, Probability & Binomial Distribution

17. The magnitude of the component of $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$ perpendicular to $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$ is:

- (a) 1
- (b) $\sqrt{11}$
- (c) $\sqrt{10}$
- (d) 3

18. If $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{j} + \hat{k}$, then $\vec{a} \times \vec{b}$ equals:

- (a) $\hat{i} + \hat{j} - \hat{k}$
- (b) $-\hat{i} + \hat{j} - \hat{k}$
- (c) $\hat{i} + \hat{j} + \hat{k}$
- (d) $\hat{i} - \hat{j} + \hat{k}$

19. The area of the parallelogram whose adjacent sides are $\vec{a} = 3\hat{i}$ and $\vec{b} = 2\hat{j}$ is:

- (a) $\sqrt{13}$
- (b) 3
- (c) 5
- (d) 6

20. If $|\vec{a}| = 4$, $|\vec{b}| = 5$ and the angle between them is 30° , then $|\vec{a} \times \vec{b}|$ equals:

- (a) 5
- (b) 20
- (c) $10\sqrt{3}$
- (d) 10

21. The area of the triangle with vertices $A(1,1,1)$, $B(2,3,4)$ and $C(4,0,2)$ is:

- (a) $\frac{\sqrt{138}}{2}$
- (b) $\frac{\sqrt{69}}{2}$
- (c) $\frac{138}{2}$
- (d) $\sqrt{138}$

22. A unit vector perpendicular to both $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{j} + \hat{k}$ is:

- (a) $\hat{i} - \hat{j} + \hat{k}$
- (b) $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$
- (c) $\frac{1}{3}(\hat{i} - \hat{j} + \hat{k})$
- (d) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

23. If $\vec{a} = 2\hat{i} + \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$, then the area of the parallelogram with diagonals $\vec{a}$ and $\vec{b}$ is:

- (a) $5\sqrt{3}$
- (b) $\frac{5\sqrt{3}}{4}$
- (c) $\frac{5\sqrt{3}}{2}$
- (d) $\sqrt{3}$

24. If $\vec{a} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{b} \neq \vec{0}$, then which of the following must hold?

- (a) $\vec{a} \cdot \vec{b} = 0$
- (b) $\vec{a} - \vec{c} = t\vec{b}$ for some scalar t
- (c) $\vec{a} = \vec{c}$

(d) $\vec{a} \perp \vec{c}$

25. The value of $[\hat{i}\hat{j}\hat{k}]$ (the scalar triple product) is:

- (a) 0
- (b) 1
- (c) 3
- (d) -1

26. The volume of the parallelepiped with edges $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$ and $\vec{c} = \hat{k} + \hat{i}$ is:

- (a) 1
- (b) 0
- (c) 3
- (d) 2

27. The value of $[\vec{a}\vec{b}\vec{c}]$ for $\vec{a} = 2\hat{i} - 3\hat{j}$, $\vec{b} = \hat{i} + \hat{j} - \hat{k}$, $\vec{c} = 3\hat{i} - \hat{k}$ is:

- (a) -4
- (b) 4
- (c) 8
- (d) 0

28. For what value of λ are the vectors $\hat{i} + \hat{j} + \hat{k}$, $2\hat{i} + 3\hat{j} + \hat{k}$ and $\hat{i} + \lambda\hat{j} - \hat{k}$ coplanar?

- (a) 1
- (b) 3
- (c) 0
- (d) -3

29. If $[\vec{a}\vec{b}\vec{c}] = 5$, then $[\vec{a} + \vec{b}\vec{b} + \vec{c}\vec{c} + \vec{a}]$ equals:

- (a) 10
- (b) 25
- (c) 15
- (d) 5

30. The volume of the parallelepiped with coterminal edges $\vec{AB}$, $\vec{AC}$, $\vec{AD}$ where $A(2,1,0)$, $B(3,3,2)$, $C(1,2,3)$ and $D(4,4,6)$ is:

- (a) 7
- (b) 0
- (c) 11
- (d) 22

31. If the vectors $2\hat{i} + 4\hat{j} + \lambda\hat{k}$ and $3\hat{i} + 6\hat{j} + 9\hat{k}$ are parallel, then λ equals:

- (a) 6
- (b) 4
- (c) 9
- (d) 3

32. The value of μ for which $\vec{a} = \mu\hat{i} + 2\hat{j} + \hat{k}$ is perpendicular to $\vec{b} = 4\hat{i} - 9\hat{j} + 2\hat{k}$ is:

- (a) 4
- (b) 8
- (c) 9
- (d) -4

33. If $\vec{a} + \vec{b}$ is perpendicular to $\vec{a} - \vec{b}$, then which of the following is true?

NDA Mathematics — Vectors, Probability & Binomial Distribution

- (a) $\vec{a} \perp \vec{b}$
 (b) $\vec{a} \cdot \vec{b} = 1$
 (c) $\vec{a} = \vec{b}$
 (d) $|\vec{a}| = |\vec{b}|$

34. If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$, then a value of λ for which $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{a}$ is:

- (a) $-\frac{14}{3}$
 (b) -14
 (c) $-\frac{3}{14}$
 (d) $\frac{14}{3}$

35. The vectors $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + p\hat{j} - 3\hat{k}$ are perpendicular when p equals:

- (a) 1
 (b) -1
 (c) -5
 (d) 5

36. The direction cosines of the vector $2\hat{i} - \hat{j} + 2\hat{k}$ are:

- (a) $\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}$
 (b) $\frac{2}{9}, -\frac{1}{9}, \frac{2}{9}$
 (c) $\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}$
 (d) 2, -1, 2

37. If l, m, n are the direction cosines of a line and $l = \frac{1}{2}$, $m = \frac{1}{2}$, then a possible value of n is:

- (a) $\frac{1}{\sqrt{2}}$
 (b) $\frac{1}{2}$
 (c) 0
 (d) 1

38. If a vector makes equal acute angles α with the three coordinate axes, then $\cos \alpha$ equals:

- (a) $\frac{1}{3}$
 (b) $\frac{\sqrt{3}}{2}$
 (c) $\frac{1}{\sqrt{2}}$
 (d) $\frac{1}{\sqrt{3}}$

39. If a line has direction ratios 1, -2, 2 and another has direction ratios 2, 2, 1, then the angle between the lines is:

- (a) 60°
 (b) 0°
 (c) 45°
 (d) 90°

40. The position vector of the midpoint of the segment joining points with position vectors $2\hat{i} + 3\hat{j}$ and $4\hat{i} - \hat{j}$ is:

- (a) $6\hat{i} + 2\hat{j}$
 (b) $\hat{i} - 2\hat{j}$
 (c) $3\hat{i} - \hat{j}$
 (d) $3\hat{i} + \hat{j}$

41. The point dividing the segment joining $A(\hat{i} + 2\hat{j} + 3\hat{k})$ and $B(4\hat{i} + 5\hat{j} + 6\hat{k})$ internally in the ratio 2: 1 has position vector:

- (a) $5\hat{i} + 7\hat{j} + 9\hat{k}$
 (b) $3\hat{i} + 4\hat{j} + 5\hat{k}$
 (c) $2\hat{i} + 3\hat{j} + 4\hat{k}$
 (d) $\frac{5}{2}\hat{i} + \frac{7}{2}\hat{j} + \frac{9}{2}\hat{k}$

42. If the points with position vectors $\hat{i} + \hat{j}$, $2\hat{i} + 3\hat{j}$ and $3\hat{i} + \lambda\hat{j}$ are collinear, then λ equals:

- (a) 4
 (b) 6
 (c) 5
 (d) 3

43. The position vector of the centroid of the triangle with vertices having position vectors $3\hat{i} - \hat{j} + 2\hat{k}$, $\hat{i} + 2\hat{j} - \hat{k}$ and $2\hat{i} + 2\hat{j} + 5\hat{k}$ is:

- (a) $2\hat{i} + \hat{j} + 2\hat{k}$
 (b) $2\hat{i} + 2\hat{j} + 2\hat{k}$
 (c) $6\hat{i} + 3\hat{j} + 6\hat{k}$
 (d) $\hat{i} + \hat{j} + 2\hat{k}$

44. The point C divides the segment joining $A(2\hat{i} + \hat{j} - \hat{k})$ and $B(8\hat{i} + 4\hat{j} - \hat{k})$ such that $\vec{C} = 4\hat{i} + 2\hat{j} - \hat{k}$. The ratio $AC:CB$ is:

- (a) 1: 1
 (b) 2: 1
 (c) 1: 3
 (d) 1: 2

45. A unit vector along the resultant of $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$ is:

- (a) $\frac{1}{\sqrt{11}}(3\hat{i} + \hat{j} - \hat{k})$
 (b) $\frac{1}{11}(3\hat{i} + \hat{j} + \hat{k})$
 (c) $3\hat{i} + \hat{j} + \hat{k}$
 (d) $\frac{1}{\sqrt{11}}(3\hat{i} + \hat{j} + \hat{k})$

46. A vector along the internal bisector of the angle between $\vec{a} = 3\hat{i} + 4\hat{j}$ and $\vec{b} = 5\hat{j}$ is proportional to:

- (a) $\hat{i} + 3\hat{j}$
 (b) $3\hat{i} + 9\hat{j}$
 (c) $8\hat{i} + 9\hat{j}$
 (d) $3\hat{i} + 4\hat{j}$

47. If $\vec{c} = 3\hat{i} + 5\hat{j}$ is written as $\vec{c} = x\vec{a} + y\vec{b}$ where $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{i} - \hat{j}$, then (x, y) is:

- (a) (-1, 4)
 (b) (4, 1)
 (c) (1, 4)
 (d) (4, -1)

48. If $\vec{a}$ and $\vec{b}$ are non-zero, non-collinear vectors and $x\vec{a} + y\vec{b} = \vec{0}$, then:

- (a) $x = y$
 (b) $x = -y$
 (c) $x = 0$ and $y = 0$

NDA Mathematics — Vectors, Probability & Binomial Distribution

(d) $x = 1, y = 1$

49. The work done by a force $\vec{F} = 2\hat{i} + 3\hat{j} + \hat{k}$ in moving a particle from the point (1,0,1) to (2,2,3) is:

- (a) 12
- (b) 6
- (c) 10
- (d) 8

50. The moment (torque) of a force $\vec{F} = 2\hat{i} - \hat{j} + 3\hat{k}$ about the origin, acting at the point with position vector $\vec{r} = \hat{i} + \hat{j} + \hat{k}$, is:

- (a) $4\hat{i} + \hat{j} - 3\hat{k}$
- (b) $2\hat{i} - \hat{j} + 3\hat{k}$
- (c) $4\hat{i} - \hat{j} - 3\hat{k}$
- (d) $-4\hat{i} + \hat{j} + 3\hat{k}$

51. Two fair dice are thrown. What is the probability that the sum of the numbers shown is 9?

- (a) $\frac{1}{9}$
- (b) $\frac{1}{6}$
- (c) $\frac{1}{12}$
- (d) $\frac{5}{36}$

52. A fair die is rolled once. What is the probability of getting a number greater than 4?

- (a) $\frac{1}{6}$
- (b) $\frac{1}{3}$
- (c) $\frac{2}{3}$
- (d) $\frac{1}{2}$

53. Three fair coins are tossed together. What is the probability of getting exactly two heads?

- (a) $\frac{3}{8}$
- (b) $\frac{1}{4}$
- (c) $\frac{1}{8}$
- (d) $\frac{1}{2}$

54. Four fair coins are tossed. What is the probability of getting at least three heads?

- (a) $\frac{1}{16}$
- (b) $\frac{1}{4}$
- (c) $\frac{3}{16}$
- (d) $\frac{5}{16}$

55. A card is drawn at random from a well-shuffled pack of 52 cards. What is the probability that it is a king?

- (a) $\frac{4}{13}$
- (b) $\frac{1}{52}$
- (c) $\frac{1}{13}$
- (d) $\frac{1}{26}$

56. A card is drawn from a pack of 52 cards. What is the probability that it is either a heart or a face card?

- (a) $\frac{3}{13}$
- (b) $\frac{1}{2}$
- (c) $\frac{25}{52}$
- (d) $\frac{11}{26}$

57. Two cards are drawn at random without replacement from a pack of 52 cards. What is the probability that both are aces?

- (a) $\frac{4}{663}$
- (b) $\frac{1}{169}$
- (c) $\frac{1}{221}$
- (d) $\frac{2}{221}$

58. Two fair dice are thrown. What is the probability that the product of the two numbers is even?

- (a) $\frac{5}{9}$
- (b) $\frac{3}{4}$
- (c) $\frac{2}{3}$
- (d) $\frac{1}{2}$

59. Three fair dice are thrown together. What is the probability that all three show different numbers?

- (a) $\frac{5}{12}$
- (b) $\frac{1}{2}$
- (c) $\frac{5}{9}$
- (d) $\frac{2}{3}$

60. Two fair dice are thrown. Given that the sum of the numbers is even, what is the probability that both numbers are themselves even?

- (a) $\frac{2}{3}$
- (b) $\frac{1}{2}$
- (c) $\frac{1}{4}$
- (d) $\frac{1}{3}$

61. A bag contains 5 red and 3 blue balls. One ball is drawn at random. What is the probability that it is blue?

- (a) $\frac{1}{3}$
- (b) $\frac{3}{5}$
- (c) $\frac{3}{8}$
- (d) $\frac{5}{8}$

62. A bag contains 4 white and 6 black balls. Two balls are drawn at random without replacement. What is the probability that both are black?

- (a) $\frac{2}{15}$
- (b) $\frac{1}{3}$

NDA Mathematics — Vectors, Probability & Binomial Distribution

- (c) $\frac{3}{5}$
(d) $\frac{9}{25}$

63. From a group of 4 men and 5 women, a committee of 3 is chosen at random. What is the probability that the committee has exactly 2 women?

- (a) $\frac{10}{21}$
(b) $\frac{2}{7}$
(c) $\frac{5}{14}$
(d) $\frac{4}{9}$

64. A box has 3 red, 4 green and 5 yellow balls. Three balls are drawn at random. What is the probability that all three are of different colours?

- (a) $\frac{3}{11}$
(b) $\frac{5}{22}$
(c) $\frac{2}{11}$
(d) $\frac{1}{4}$

65. The letters of the word INDIA are arranged at random in a row. What is the probability that the two I's come together?

- (a) $\frac{1}{10}$
(b) $\frac{1}{5}$
(c) $\frac{3}{5}$
(d) $\frac{2}{5}$

66. Out of 7 consonants and 4 vowels, a group of 3 consonants and 2 vowels is to be selected. What is the probability of selecting a specific particular set of 3 consonants and 2 vowels?

- (a) $\frac{1}{35}$
(b) $\frac{1}{462}$
(c) $\frac{1}{6}$
(d) $\frac{1}{210}$

67. Five persons including A and B are seated at random in a row. What is the probability that A and B sit next to each other?

- (a) $\frac{1}{5}$
(b) $\frac{3}{5}$
(c) $\frac{1}{10}$
(d) $\frac{2}{5}$

68. For two events, $P(A) = 0.5$, $P(B) = 0.3$ and $P(A \cap B) = 0.1$. Find $P(A \cup B)$.

- (a) $\frac{3}{5}$
(b) $\frac{9}{10}$
(c) $\frac{4}{5}$
(d) $\frac{7}{10}$

69. If A and B are mutually exclusive events with $P(A) = \frac{1}{4}$ and $P(B) = \frac{1}{3}$, find $P(A \cup B)$.

- (a) $\frac{7}{12}$
(b) $\frac{1}{7}$
(c) $\frac{1}{12}$
(d) $\frac{5}{12}$

70. A number is chosen at random from 1 to 30. What is the probability that it is divisible by 3 or by 5?

- (a) $\frac{7}{15}$
(b) $\frac{8}{15}$
(c) $\frac{1}{2}$
(d) $\frac{2}{5}$

71. If $P(A) = \frac{2}{5}$, $P(B) = \frac{1}{3}$ and $P(A \cup B) = \frac{3}{5}$, find $P(A \cap B)$.

- (a) $\frac{1}{5}$
(b) $\frac{1}{15}$
(c) $\frac{2}{15}$
(d) $\frac{4}{15}$

72. A card is drawn from a pack of 52. What is the probability that it is a spade or a king or a red queen?

- (a) $\frac{9}{26}$
(b) $\frac{4}{13}$
(c) $\frac{1}{4}$
(d) $\frac{5}{13}$

73. The probability that it will rain tomorrow is $\frac{2}{7}$. What is the probability that it will not rain tomorrow?

- (a) $\frac{2}{7}$
(b) $\frac{4}{7}$
(c) $\frac{5}{7}$
(d) $\frac{1}{7}$

74. Three coins are tossed. What is the probability of getting at least one tail?

- (a) $\frac{3}{8}$
(b) $\frac{1}{8}$
(c) $\frac{7}{8}$
(d) $\frac{1}{2}$

75. A bag contains 6 red and 4 green balls. Two balls are drawn at random without replacement. What is the probability of getting at least one green ball?

- (a) $\frac{2}{3}$
(b) $\frac{8}{15}$

NDA Mathematics — Vectors, Probability & Binomial Distribution

- (c) $\frac{3}{5}$
(d) $\frac{1}{3}$

76. A fair die is thrown 3 times. What is the probability of getting at least one six?

- (a) $\frac{91}{216}$
(b) $\frac{125}{216}$
(c) $\frac{25}{216}$
(d) $\frac{1}{2}$

77. The probabilities that a problem is solved by three students independently are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. What is the probability that the problem is solved by at least one of them?

- (a) $\frac{2}{3}$
(b) $\frac{1}{24}$
(c) $\frac{3}{4}$
(d) $\frac{1}{4}$

78. If $P(A \cap B) = \frac{1}{4}$ and $P(B) = \frac{1}{2}$, find $P(A|B)$.

- (a) $\frac{1}{2}$
(b) $\frac{3}{4}$
(c) $\frac{1}{4}$
(d) $\frac{1}{8}$

79. A fair die is rolled. Given that the number obtained is even, what is the probability that it is greater than 3?

- (a) $\frac{1}{2}$
(b) $\frac{1}{3}$
(c) $\frac{1}{6}$
(d) $\frac{2}{3}$

80. Two cards are drawn one after another without replacement from a pack of 52. Given that the first card is a king, what is the probability that the second card is also a king?

- (a) $\frac{1}{13}$
(b) $\frac{3}{52}$
(c) $\frac{4}{51}$
(d) $\frac{1}{17}$

81. Two dice are thrown. Given that the sum of the numbers is 7, what is the probability that one of the dice shows a 2?

- (a) $\frac{1}{2}$
(b) $\frac{1}{6}$
(c) $\frac{1}{3}$
(d) $\frac{2}{3}$

82. A family has two children. Given that at least one of them is a boy, what is the probability that both are boys? (Assume boys and girls are equally likely.)

- (a) $\frac{2}{3}$
(b) $\frac{1}{3}$
(c) $\frac{1}{2}$
(d) $\frac{1}{4}$

83. If $P(A) = \frac{3}{8}$, $P(B) = \frac{5}{8}$ and $P(A \cup B) = \frac{3}{4}$, find $P(A|B)$.

- (a) $\frac{1}{5}$
(b) $\frac{1}{4}$
(c) $\frac{3}{5}$
(d) $\frac{2}{5}$

84. A and B are independent events with $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{4}$. Find $P(A \cap B)$.

- (a) $\frac{5}{12}$
(b) $\frac{1}{12}$
(c) $\frac{7}{12}$
(d) $\frac{1}{7}$

85. The probabilities that two independent switches A and B work are 0.8 and 0.6 respectively. What is the probability that both work?

- (a) $\frac{14}{25}$
(b) $\frac{1}{5}$
(c) $\frac{12}{25}$
(d) $\frac{7}{10}$

86. A coin is tossed and a die is rolled. What is the probability of getting a head and a number greater than 4?

- (a) $\frac{1}{6}$
(b) $\frac{1}{3}$
(c) $\frac{1}{12}$
(d) $\frac{1}{2}$

87. Two independent events A and B have $P(A) = \frac{2}{5}$ and $P(B) = \frac{1}{2}$. Find $P(A \cup B)$.

- (a) $\frac{9}{10}$
(b) $\frac{7}{10}$
(c) $\frac{1}{5}$
(d) $\frac{3}{5}$

88. A problem is given to three students whose chances of solving it independently are $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{5}$. What is the probability that exactly one of them solves it?

- (a) $\frac{3}{5}$

NDA Mathematics — Vectors, Probability & Binomial Distribution

- (b) $\frac{17}{30}$
 (c) $\frac{13}{30}$
 (d) $\frac{1}{60}$

89. A and B are independent events such that $P(A) = 0.3$ and $P(A \cup B) = 0.65$. Find $P(B)$.

- (a) $\frac{1}{2}$
 (b) $\frac{2}{5}$
 (c) $\frac{7}{20}$
 (d) $\frac{3}{5}$

90. Bag I contains 3 red and 2 black balls; Bag II contains 2 red and 4 black balls. A bag is chosen at random and a ball is drawn. What is the probability that the ball drawn is red?

- (a) $\frac{7}{15}$
 (b) $\frac{2}{5}$
 (c) $\frac{1}{2}$
 (d) $\frac{8}{15}$

91. Bag I has 3 red and 2 black balls, Bag II has 2 red and 4 black balls. A bag is chosen at random and a red ball is drawn. What is the probability that it came from Bag I?

- (a) $\frac{5}{14}$
 (b) $\frac{9}{14}$
 (c) $\frac{1}{2}$
 (d) $\frac{3}{7}$

92. In a factory, machines A and B produce 60% and 40% of the items. 2% of A's output and 5% of B's output are defective. What is the probability that a randomly chosen item is defective?

- (a) $\frac{7}{250}$
 (b) $\frac{1}{25}$
 (c) $\frac{3}{100}$
 (d) $\frac{4}{125}$

93. Machines A and B produce 60% and 40% of items; 2% of A's and 5% of B's are defective. If a chosen item is defective, what is the probability that it was made by machine B?

- (a) $\frac{2}{5}$
 (b) $\frac{5}{8}$
 (c) $\frac{1}{2}$
 (d) $\frac{3}{8}$

94. A box contains 2 gold and 3 silver coins; a second box contains 4 gold and 1 silver coin. A box is selected at random and a gold coin is drawn. What is the probability that it came from the second box?

- (a) $\frac{3}{5}$
 (b) $\frac{2}{3}$
 (c) $\frac{1}{3}$
 (d) $\frac{4}{5}$

95. A test detects a disease correctly 99% of the time when present and gives a false positive 2% of the time. If 1% of the population has the disease, what is the probability that a person who tests positive actually has the disease?

- (a) $\frac{1}{2}$
 (b) $\frac{2}{3}$
 (c) $\frac{99}{100}$
 (d) $\frac{1}{3}$

96. The odds in favour of an event are 3: 5. What is the probability that the event occurs?

- (a) $\frac{3}{5}$
 (b) $\frac{5}{8}$
 (c) $\frac{5}{3}$
 (d) $\frac{3}{8}$

97. If the probability of an event is $\frac{2}{7}$, what are the odds against the event?

- (a) 2: 7
 (b) 5: 2
 (c) 5: 7
 (d) 2: 5

98. What is the probability that a leap year selected at random will contain 53 Sundays?

- (a) $\frac{3}{7}$
 (b) $\frac{53}{366}$
 (c) $\frac{1}{7}$
 (d) $\frac{2}{7}$

99. What is the probability that a non-leap year selected at random will contain 53 Mondays?

- (a) $\frac{3}{7}$
 (b) $\frac{1}{7}$
 (c) $\frac{2}{7}$
 (d) $\frac{53}{365}$

100. Three persons are chosen at random. What is the probability that all three were born on different days of the week? (Assume each day equally likely.)

- (a) $\frac{1}{7}$
 (b) $\frac{6}{7}$
 (c) $\frac{30}{49}$
 (d) $\frac{19}{49}$

NDA Mathematics — Vectors, Probability & Binomial Distribution

101. A fair coin is tossed 5 times. What is the probability of getting exactly 3 heads?

- (a) $\frac{5}{16}$
- (b) $\frac{10}{16}$
- (c) $\frac{3}{16}$
- (d) $\frac{1}{2}$

102. A die is thrown 4 times. What is the probability of getting exactly two sixes?

- (a) $\frac{25}{1296}$
- (b) $\frac{25}{216}$
- (c) $\frac{5}{216}$
- (d) $\frac{1}{36}$

103. In a box 20% of the bulbs are defective. If 5 bulbs are drawn at random (with replacement), what is the probability that exactly 2 are defective?

- (a) $\left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^3$
- (b) $\left(\frac{5}{2}\right) \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^3$
- (c) $\left(\frac{5}{2}\right) \left(\frac{1}{5}\right)^2$
- (d) $\left(\frac{5}{2}\right) \left(\frac{4}{5}\right)^2 \left(\frac{1}{5}\right)^3$

104. A marksman hits a target with probability $\frac{3}{4}$. If he fires 4 shots, what is the probability that he hits the target exactly 3 times?

- (a) $\frac{81}{256}$
- (b) $\frac{27}{256}$
- (c) $\frac{9}{64}$
- (d) $\frac{27}{64}$

105. The probability that a seed germinates is 0.6. If 6 seeds are sown, what is the probability that exactly 4 germinate?

- (a) $\frac{4860}{15625}$
- (b) $\frac{3240}{15625}$
- (c) $\frac{4860}{46875}$
- (d) $\frac{1296}{15625}$

106. A fair die is thrown 18 times. What is the mean number of sixes obtained?

- (a) 9
- (b) 6
- (c) 3
- (d) $\frac{5}{2}$

107. For a binomial distribution with $n = 12$ and $p = \frac{1}{4}$, what is the variance?

- (a) $\frac{27}{16}$
- (b) $\frac{3}{4}$
- (c) 3

(d) $\frac{9}{4}$

108. In a binomial distribution the mean is 4 and the variance is $\frac{4}{3}$. What is the number of trials n ?

- (a) 12
- (b) 4
- (c) 8
- (d) 6

109. The mean and variance of a binomial distribution are 6 and 4 respectively. What is the value of p ?

- (a) $\frac{1}{3}$
- (b) $\frac{1}{2}$
- (c) $\frac{2}{3}$
- (d) $\frac{1}{6}$

110. For a binomial distribution the mean is 20 and the standard deviation is 4. What is the number of trials n ?

- (a) 100
- (b) 80
- (c) 25
- (d) 50

111. A die is thrown 4 times. What is the probability of getting at least one six?

- (a) $\frac{4}{6}$
- (b) $\frac{1}{6}$
- (c) $\frac{671}{1296}$
- (d) $\frac{625}{1296}$

112. A fair coin is tossed 3 times. What is the probability of getting at least one head?

- (a) $\frac{1}{8}$
- (b) $\frac{7}{8}$
- (c) $\frac{3}{8}$
- (d) $\frac{1}{2}$

113. The probability that a bulb is defective is $\frac{1}{10}$. In a sample of 3 bulbs, what is the probability that at most one is defective?

- (a) $\frac{28}{1000}$
- (b) $\frac{729}{1000}$
- (c) $\frac{972}{1000}$
- (d) $\frac{243}{1000}$

114. A marksman hits a target with probability $\frac{1}{3}$. If he fires 5 shots, what is the probability that he hits the target at least twice?

- (a) $\frac{112}{243}$
- (b) $\frac{32}{243}$
- (c) $\frac{80}{243}$

NDA Mathematics — Vectors, Probability & Binomial Distribution

(d) $\frac{131}{243}$

115. A fair coin is tossed 7 times. What is the most probable number of heads?

- (a) 4 only
- (b) 3 or 4
- (c) 2 or 5
- (d) 3 only

116. A die is thrown 10 times. What is the most probable number of sixes?

- (a) 2
- (b) 3
- (c) 1
- (d) 0

117. For a binomial distribution with $n = 9$ and $p = \frac{1}{3}$, what is the most probable number of successes?

- (a) 2 or 3
- (b) 2
- (c) 3
- (d) 4

118. What is the least number of times a fair coin must be tossed so that the probability of getting at least one head exceeds 0.9?

- (a) 4
- (b) 5
- (c) 3
- (d) 10

119. For a binomial distribution with $n = 6$ and $p = \frac{1}{3}$, what is the ratio $\frac{P(X=2)}{P(X=1)}$?

- (a) $\frac{5}{4}$
- (b) $\frac{4}{5}$
- (c) $\frac{15}{8}$
- (d) $\frac{5}{2}$

120. If X follows a binomial distribution with $n = 4$ and $p = \frac{1}{2}$, what is $P(X = 0) + P(X = 4)$?

- (a) $\frac{1}{16}$
- (b) $\frac{1}{2}$
- (c) $\frac{1}{4}$
- (d) $\frac{1}{8}$